

Machine Learning for Generation of Graphs of Given Degree and Girth

Bachelor's Thesis Defense

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The Problem



A (k, g) -graph is a graph that is

- **k -regular** — every vertex has degree exactly k , and
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A classical **extremal problem**: best known orders are still improved today by a mix of hand constructions and large computer searches.

Why ML cannot be a black box here

Two defining constraints, two very different difficulties:

- **Degree** is **local** — a one-hop count.
- **Girth** is **global** — needs paths that leave and return.

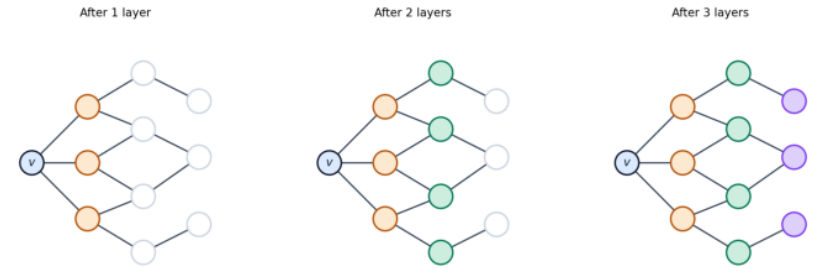
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Hard limits:

- Message-passing GNN expressivity $\leq$ 1-WL.
- **Girth is provably not computable by an MP-GNN** (Garg et al.).
- RL only works where the move space already carries structure.



Depth = how far information travels;
a global property like girth slips past
a fixed receptive field.

Not “**produce a cage.**”

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A learned component **ranks / proposes**; an exact check **accepts**.

Diagnostic: What Can a GNN Learn?

Degree is learnable, girth is not

Probe the two defining constraints directly as per-node predictions (trained on fresh random graphs, exact labels):

- **Degree — solved:** SAGE with **4,900 params** reaches **accuracy 1.000**.
- **Min-cycle / girth — resists every architecture:** best is the cycle-aware **Loopy GNN at 0.510** (mislabels half).

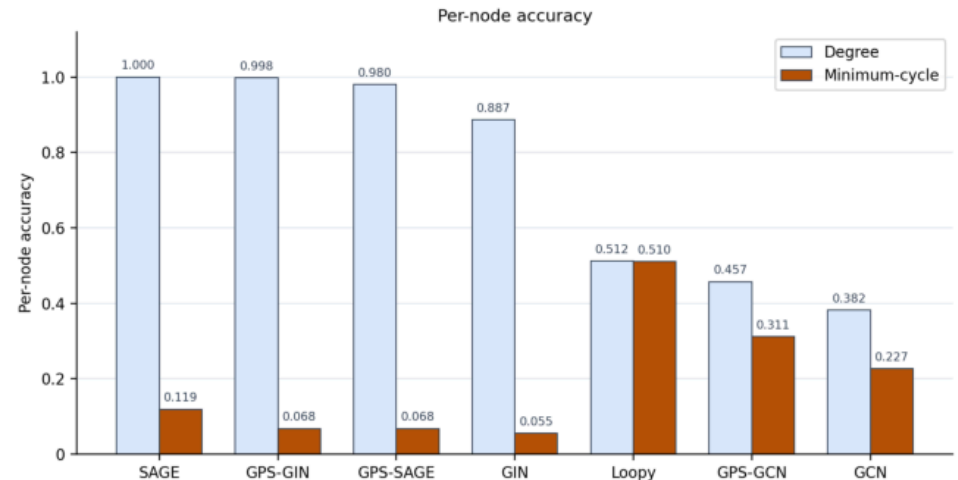
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Consequence (drives the whole design): degree gets **built into the**



The split: degree is recovered exactly;
the minimum cycle is not.

Degree is learnable, girth is not

construction; girth gets handled by
explicit exact search.

Construction Methods

A ladder of increasing structure

1. **Direct RL** — edit edges under a learned policy (*least structure*)
2. **Voltage lifts** — regularity built in, only girth searched
3. **Refinement & Excision** — repair near-misses, then shrink
4. **Forge** — compose a producer with refinement + excision (*most structure*)

1. Direct RL

State = partial graph · Action = add/remove a vertex pair · trained with **PPO**.

- Illegal moves are **pruned** (degree cap, would-be short cycle, disconnection).
- Needs a **curriculum** (start at (3, 5), unlock harder targets) — **without it the agent never even makes a (3, 5)-graph**.
- Needs a **potential-based shaped reward** to learn at all.

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Limits: regularity is not guaranteed, and the action space grows **quadratically** $\binom{n}{2} \rightarrow$ does not scale. $\rightarrow$ motivates structure.

2. Voltage lifts

Idea: **k -regularity becomes structural** — every lift of a k -regular base graph is k -regular. Only choice that affects girth: the **voltage assignment** on a small base.

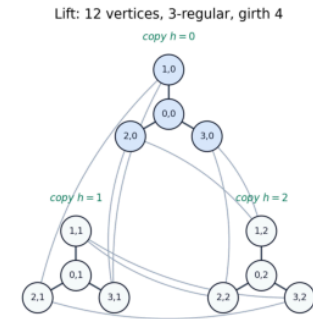
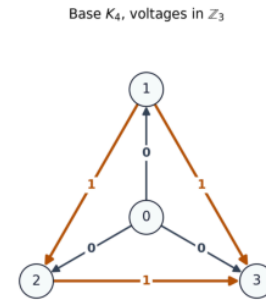
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Read girth **off the base** via net voltages:

$$\text{girth}(\text{lift}) = \min_W |W| \cdot \text{ord}(s)$$

Gauge freedom zeroes spanning-tree voltages $\rightarrow$ search only non-tree edges (e.g. K_4 : **3 voltages instead of 6**).



K_4 over $\mathbb{Z}_3$: a 12-vertex, 3-regular connected lift.

3. Refinement & Excision

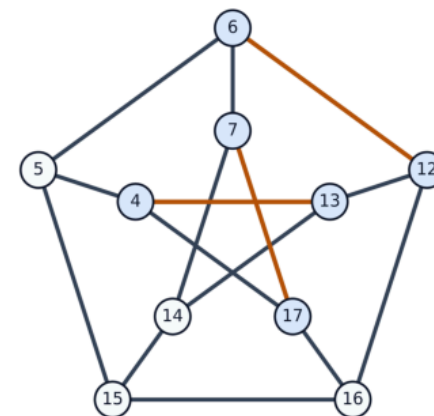
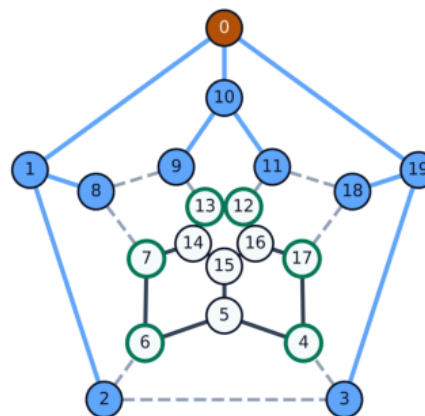
Refinement — fix a near-miss lift without discarding it: degree-preserving **2-/3-swaps** via tabu search, weighting short cycles exponentially. Learned evaluator **ranks** swaps; exact check decides.

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Excision — make a valid graph **smaller**: remove a tree of **Moore radius** $\left\lfloor \frac{g-1}{2} \right\rfloor$, then stitch deficient vertices at distance $\geq g - 1$. Learned (k, g) -independent repair policy

1. **exact backtracking** fallback.



Excision: dodecahedral graph $\rightarrow$ **Petersen graph**, the $(3, 5)$ -cage — half the order.

4. Forge

Compose the methods so each does only what it does well:

Producer

(voltage method)

→ *reach*

Refinement

rescue near-misses

→ *validity*

Excision

shrink to cage

→ *size*

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- Near-miss hand-off via the **defective fraction** $\leq \tau$ (scales with size).
- Ordering is forced; stages run **round-robin**; producer is **interchangeable**.

Results

Coverage vs. size, across 22 targets

- **voltage-rl** has by far the **widest reach** — the only method solving (3, 9), (3, 10), (4, 8), (6, 6), and (6, 5) at 57% of seeds.
- **Classical** methods are **size-optimal but narrow**; direct RL is weakest.
- **No method** solves (4, 7), (5, 7), (7, 5) within the 60 s budget.

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- **No method** solves (4, 7), (5, 7), (7, 5) within the 60 s budget.
- **Size:** the voltage family lands at **2× the Moore bound**; classical methods are smallest where they work at all.

Forge: the balanced answer

- Tabu-based producers: **0.46 success**, **1.23× Moore bound** — and **meet the cage exactly on the smallest targets**.
- The RL producer is the **strongest standalone** policy yet the **weakest inside Forge**: the pipeline wants a **diverse** stream of near-cage lifts.

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- **Ablation**: remove excision $\rightarrow$ size **1.51 $\rightarrow$ 2.06** (excision is what shrinks); remove refinement $\rightarrow$ success **0.31 $\rightarrow$ 0.62** but larger graphs.

Conclusion

What we learned

- **Degree** essentially **solved** (tiny model, perfect); **girth resists every architecture** → handled by exact search.
- Dominant pattern: a **coverage-vs-size trade-off**; **Forge is the most balanced** ($1.23\times$ Moore, keeps much of the reach).

What we learned

- **Degree** essentially **solved** (tiny model, perfect); **girth** **resists every architecture** → handled by exact search.
- Dominant pattern: a **coverage-vs-size trade-off**; **Forge** is the most balanced ($1.23\times$ Moore, keeps much of the reach).
- **The honest finding: learned components rarely beat their non-learned counterparts** — and that **is** the result.
- Learning earns a place **beside** structured search: the **learned excision repair**, and **RL as a diverse lift source** in Forge.

Learning guides where the search looks; exact algorithms decide validity.

- **Co-adapt** Forge's three stages (currently tuned in isolation).
- **Learn** the excision root and depth; train the producer for **diversity**.
- A **girth-specialised, degree-agnostic** model (degree is free under lifts).
- Ongoing parallel **record search** on open targets $(8, 5)$, $(9, 5)$, $(10, 5)$, $(11, 6)$.

Thank you — questions welcome.

Backup



Girth on the base graph $(\frac{K_4}{\mathbb{Z}_3})$

Net voltage of a reduced closed walk

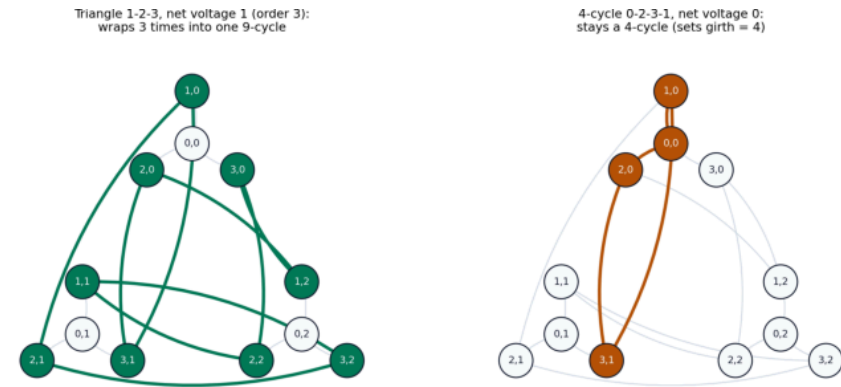
W :

- $\text{net} = \text{identity} \rightarrow \text{cycle of length } |W|$;
- $\text{else} \rightarrow \text{cycle of length } |W| \cdot \text{ord}(s)$.

$$\text{girth}(\text{lift}) = \min_W |W| \cdot \text{ord}(s)$$

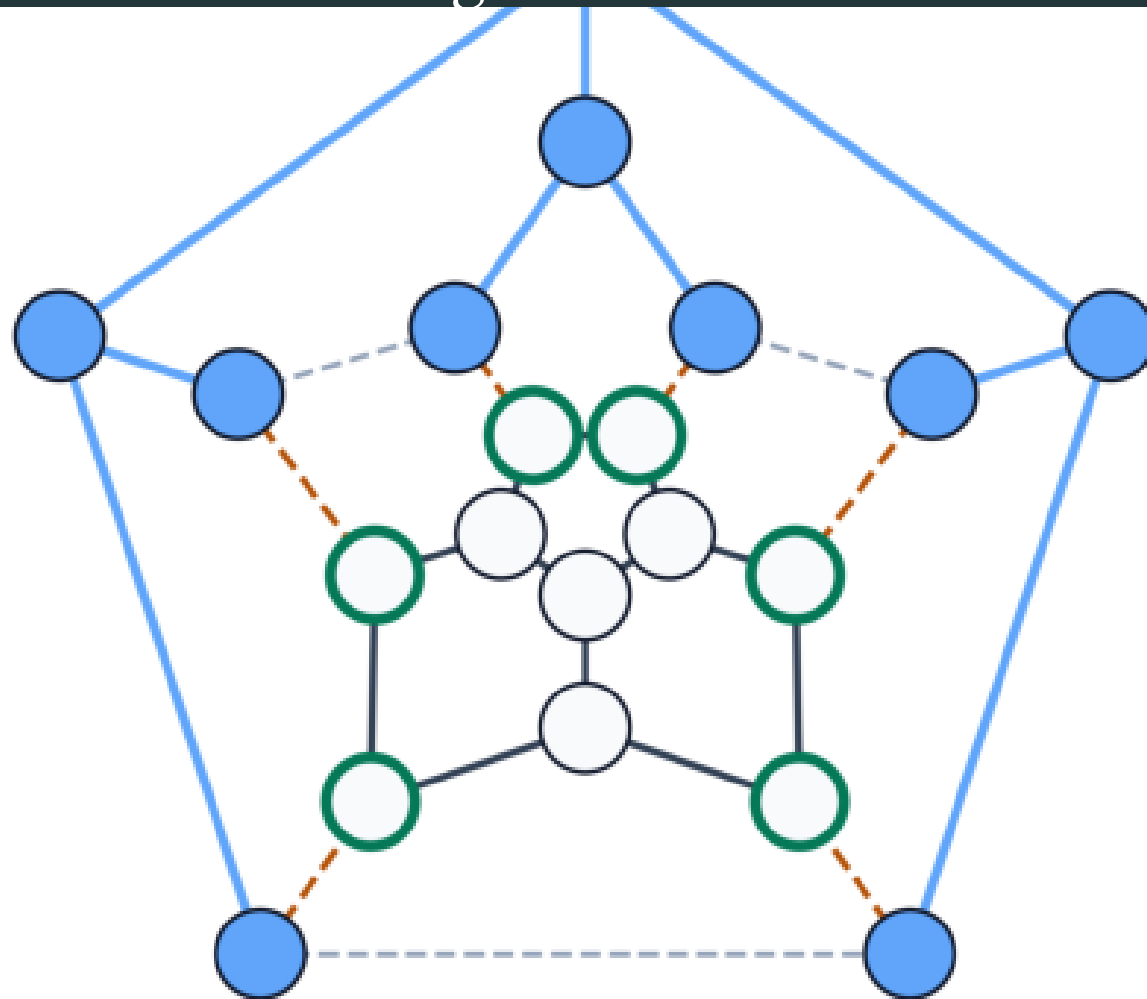
In $\frac{K_4}{\mathbb{Z}_3}$: triangles (order 3) wrap into a **9-cycle**; the 4-cycle survives $\rightarrow$ **girth 4**.

No assignment reaches girth 5.



Base cycles lift to short cycles; the surviving 4-cycle fixes girth.

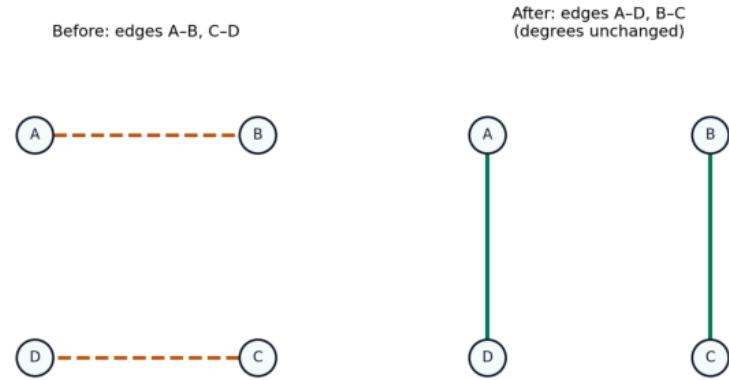
Excision: tree removal + stitching



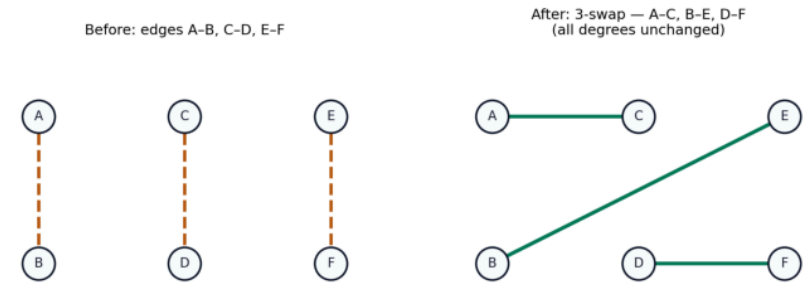
A depth-2 BFS tree removed from the dodecahedral graph leaves deficient

boundary vertices, reconnected only at distance $> q - 1$.

Degree-preserving swaps



2-swap: rewire two disjoint edges;
degrees preserved, a triangle can be
broken.

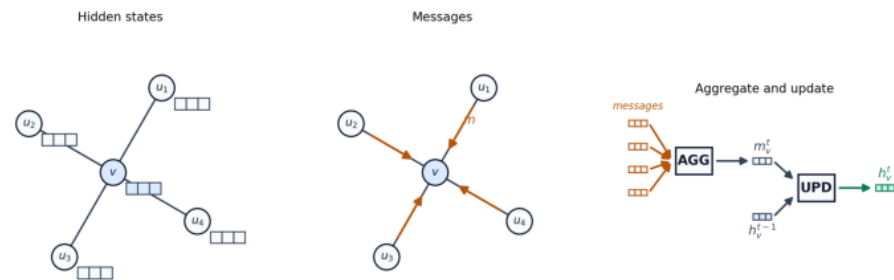


3-swap: reaches configurations no 2-
swap can.

Message passing

$$\mathbf{h}_v^{(k)} = \phi \left(\mathbf{h}_v^{(k-1)}, \bigoplus_{u \in \mathcal{N}(v)} \psi(\mathbf{h}_v^{(k-1)}, \mathbf{h}_u^{(k-1)}) \right)$$

Same local update at every vertex $\rightarrow$ size-independent and permutation-equivariant. Sum aggregation keeps counts (degree); normalized aggregation hides them.



One message-passing layer: aggregate then update.

Evaluation setup

- Trained on the **PERUN** supercomputer (TUKER).
- **60-second** budget per attempt; single node (256 cores), **128 parallel workers** (256 oversubscribes the cores → collapse).
- Architectures probed: GCN, GraphSAGE, GIN, GINE, GPS (graph transformer), Loopy (cycle-aware).
- Property predictors trained on **Erdős–Rényi** graphs (fresh per step → no memorization), with exact labels.